

DO I NEED TO DO MATH HOMEWORK?

Do times tables terrify you? Do fractions make you fearful? Does division blur your vision? Are you simply scared of subtraction? Be not afraid, for what your math teachers have been telling you for years is absolutely true: mathematics really is important in our everyday lives. It all adds up to this—without it, we'd be clueless.

WHY MATH MATTERS

Mathematics is a key tool in science. Without it, there would be no physics, chemistry, or astronomy, for example. Math help us get from point A to point B. We depend on math in navigation, aviation, and map-making. In fact, without math, any field that depends on numbers, from architecture to commerce, would fail. We couldn't even get to school in the morning without using math to stay on time. Even in the past, math was essential. It helped the world's greatest empires rise to power.

EGYPTIAN EXPERTS

The ancient Egyptians were excellent at math. They used a decimal system based on ten fingers of the hand, figured out how to multiply and divide with addition and subtraction, and understood concepts such as binary numbers, fractions, and solids. As you might expect from the people who brought you the pyramids, they knew all about triangles. Mathematicians in the ancient world also made key developments in timekeeping. Thanks to the Babylonians, we have 60 seconds in one minute and 60 minutes in one hour, not to mention 360 degrees in a circle.

GREEK GODS

The ancient Greeks were geniuses of geometry, using it to prove that certain mathematical ideas were always true. Top brains including Pythagoras (500s BCE), Aristotle (350s BCE), Euclid (300s BCE), and Archimedes (200s BCE) came up with brilliant theories that you still use in math class today.

OLD CHINA

Mathematics helped the Chinese accomplish amazing feats of engineering, such as the Great Wall. The abacus, attributed to the Chinese, is a device that can add, subtract, multiply, and divide, as well as work with sophisticated mathematical problems such as square roots. The earliest version of the game sudoku appeared in an ancient Chinese text dating back to about 1000 BCE.

INDIAN INFINITY

The decimal and binary number systems were first recorded in Indian math books. The concept of the number zero may have started in India, as well as the understanding of infinity and negative numbers. Between the years 622 and 1600, mathematics flourished in the Islamic world. The mathematicians had ideas way ahead of their time. Algebra, calculus, and trigonometry were developed there.

EUROPEAN EGGHEADS

Key European mathematicians included Leonardo Fibonacci (1200s), who described a number sequence in which each number is the sum of the previous two numbers, René Descartes (1596–1650), who found that curved lines can be described by equations, and Sir Isaac Newton (1643–1727), who made developments in calculus.

OUTNUMBERED

You might not make mathematical history like these legends did, but math is good for your brain. Practicing how to solve mathematical problems boosts your ability to make complex decisions. Learning new math concepts forces you to think in different ways so that your brain can build the connections it needs to solve all kinds of challenges. Now all that's left to say is please go do your math homework!